T.P. Nro 1: Sintaxis y semántica de los lenguajes

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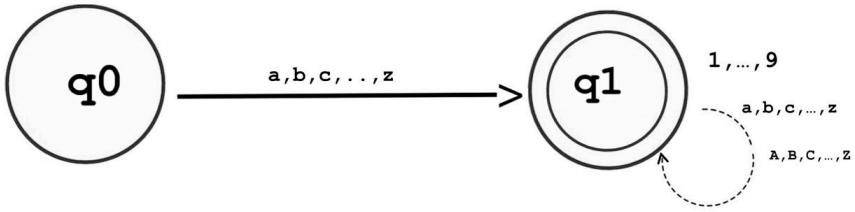
Tokens posibles:

id,num,const,var, call,begin, end, if, then, while , do, procedure , # , ; , , ,assign, comparison, operation ,*, /, (,), odd, espacio_blanco

AFDs y Expresiones Regulares:

Referencias en color ROJO para diferenciar los AFDs de sus respectivas ER. (Véase ejemplo en token “Num”)

Id:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1}

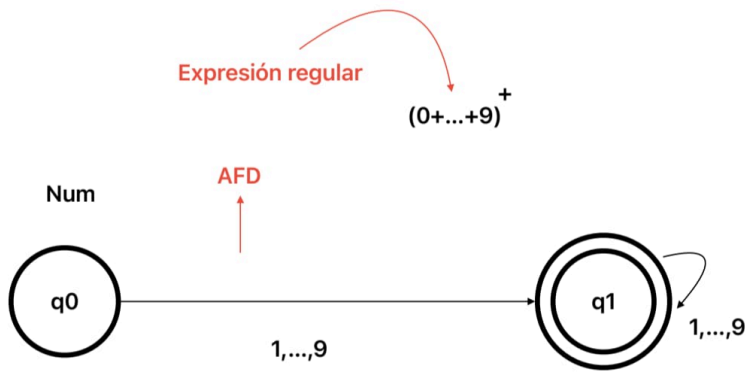
Σ= {UTF-8}

F={q1}

Q0={q0}

	a	b	c	…	z	A	B	C	…	Z	1	…	9
q0	q1	q1	q1	q1	q1	qt	qt	qt	qt	qt	qt	qt	qt
q1	q1	q1	q1	q1	q1	q1	q1	q1	q1	q1	q1	q1	q1
qt	qt	qt	qt	qt	qt	qt	qt	qt	qt	qt	qt	qt	qt

Num:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1}

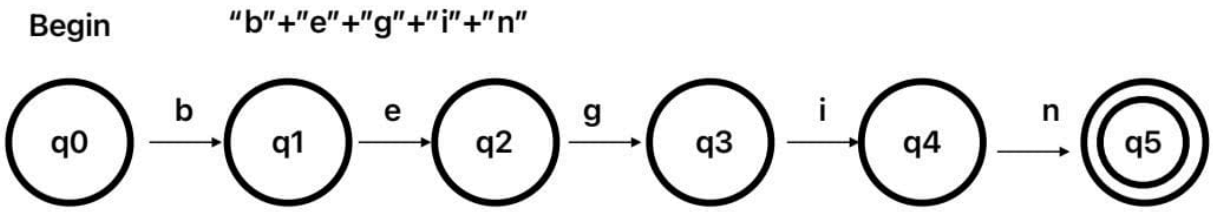
Σ= {UTF-8}

F={q1}

Q0={q0}

	1	…	9
q0	q1	q1	q1
q1	q1	q1	q1
qt	qt	qt	qt

Begin:



A=(VT,VN,Σ,F,Q0)

VT={q0,q1,q2,q3,q4,q5}

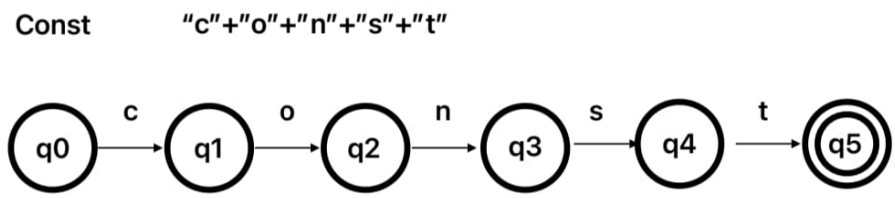
Σ= {UTF-8}

F={q5}

Q0={q0}

	b	e	g	i	n
q0	q1	qt	qt	qt	qt
q1	qt	q2	qt	qt	qt
q2	qt	qt	q3	qt	qt
q3	qt	qt	qt	q4	qt
q4	qt	qt	qt	qt	q5
q5	qt	qt	qt	qt	qt
qt	qt	qt	qt	qt	qt

Const:



A=(VT,VN,Σ,F,Q0)

VT= {q0,q1,q2,q3,q4,q5}

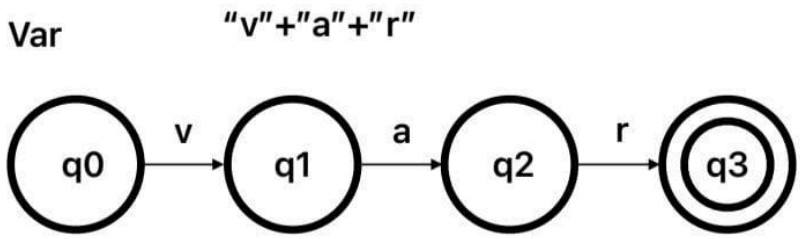
Σ= {UTF-8}

F={q5}

Q0={q0}

	c	o	n	s	t
q0	q1	qt	qt	qt	qt
q1	qt	q2	qt	qt	qt
q2	qt	qt	q3	qt	qt
q3	qt	qt	qt	q4	qt
q4	qt	qt	qt	qt	q5
q5	qt	qt	qt	qt	qt
qt	qt	qt	qt	qt	qt

Var:



A=(VT,VN,Σ,F,Q0)

VT= {q0,q1,q2,q3}

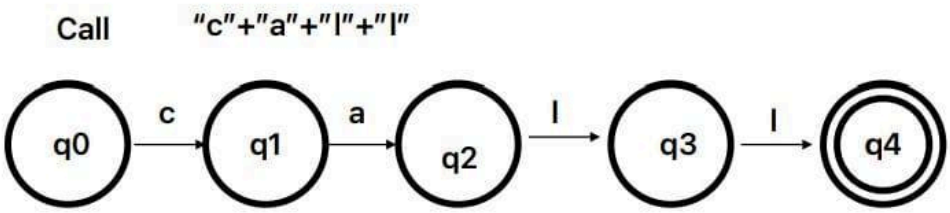
Σ= {UTF-8}

F={q3}

Q0={q0}

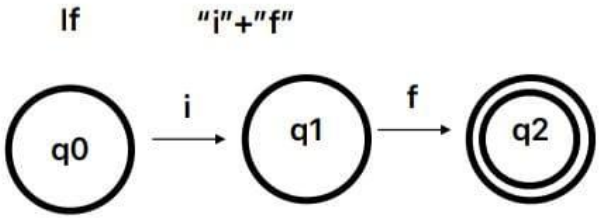
	v	a	r
q0	q1	qt	qt
q1	qt	q2	qt
q2	qt	qt	q3
q3	qt	qt	qt
qt	qt	qt	qt

Call:



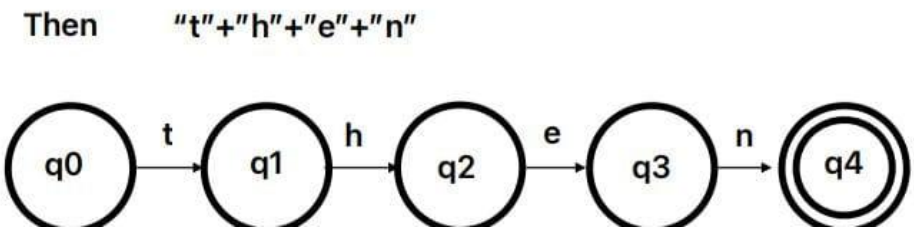
A=(VT,VN,Σ,F,Q0)		c	a	l	l
VT= {q0,q1,q2,q3,q4}	q0	q1	qt	qt	qt
Σ= {UTF-8}	q1	qt	q2	qt	qt
F={q4}	q2	qt	qt	q3	qt
Q0={q0}	q3	qt	qt	qt	q4
	q4	qt	qt	qt	qt
	qt	qt	qt	qt	qt

If:



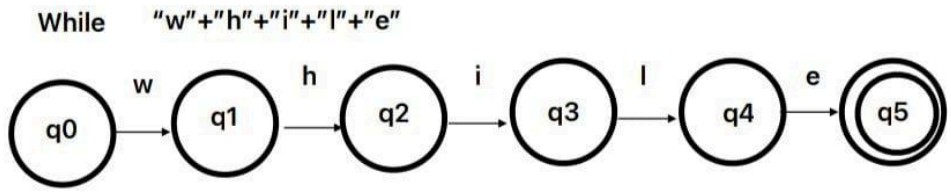
A=(VT,VN,Σ,F,Q0)		i	f
VT= {q0,q1,q2}	q0	q1	qt
Σ= {UTF-8}	q1	qt	q2
F={q2}	q2	qt	qt
Q0={q0}	qt	qt	qt

Then:



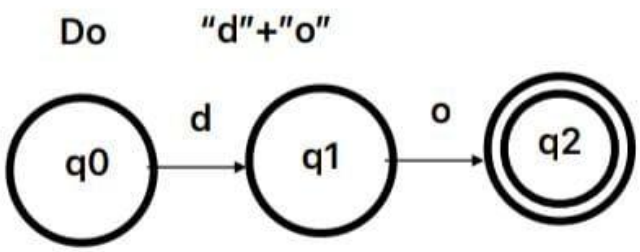
A=(VT,VN,Σ,F,Q0)		t	h	e	n
VT= {q0,q1,q2,q3,q4}	q0	q1	qt	qt	qt
Σ= {UTF-8}	q1	qt	q2	qt	qt
F={q4}	q2	qt	qt	q3	qt
Q0={q0}	q3	qt	qt	qt	q4
	q4	qt	qt	qt	qt
	qt	qt	qt	qt	qt

While:



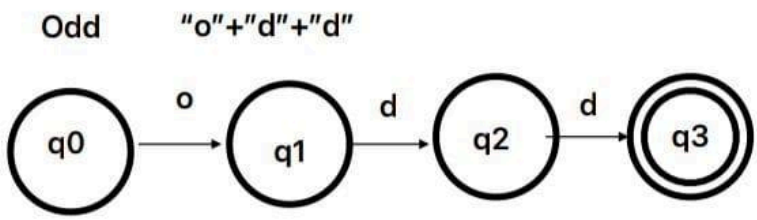
A=(VT,VN,Σ,F,Q0)		w	h	i	l	e
VT= {q0,q1,q2,q3,q4,q5}	q0	q1	qt	qt	qt	qt
Σ= {UTF-8}	q1	qt	q2	qt	qt	qt
F={q5}	q2	qt	qt	q3	qt	qt
Q0={q0}	q3	qt	qt	qt	q4	qt
	q4	qt	qt	qt	qt	q5
	q5	qt	qt	qt	qt	qt
	qt	qt	qt	qt	qt	qt

Do:



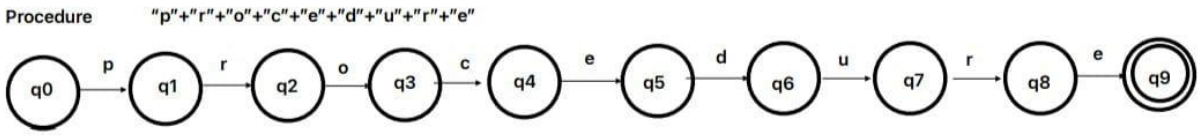
A=(VT,VN,Σ,F,Q0)		d	o
VN= {q0,q1,q2}	q0	q1	qt
Σ= {UTF-8}	q1	qt	q2
F={q2}	q2	qt	qt
Q0={q0}	qt	qt	qt

Odd:



A=(VT,VN,Σ,F,Q0)		o	d	d
VN= {q0,q1,q2,q3}	q0	q1	qt	qt
Σ= {UTF-8}	q1	qt	q2	qt
F={q3}	q2	qt	qt	q3
Q0={q0}	q3	qt	qt	qt
	qt	qt	qt	qt

Procedure:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1,q2,q3,q4,q5,q6,q7,q8,q9}

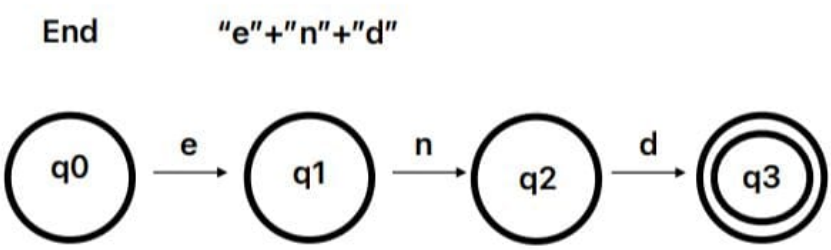
Σ= {UTF-8}

F={q9}

Q0={q0}

	p	r	o	c	e	d	u	r	e
q0	**q1**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**
q1	**qt**	**q2**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**
q2	**qt**	**qt**	**q3**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**
q3	**qt**	**qt**	**qt**	**q4**	**qt**	**qt**	**qt**	**qt**	**qt**
q4	**qt**	**qt**	**qt**	**qt**	**q5**	**qt**	**qt**	**qt**	**qt**
q5	**qt**	**qt**	**qt**	**qt**	**qt**	**q6**	**qt**	**qt**	**qt**
q6	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**q7**	**qt**	**qt**
q7	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**q8**	**qt**
q8	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**q9**
q9	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**
qt	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**	**qt**

End:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1,q2,q3}

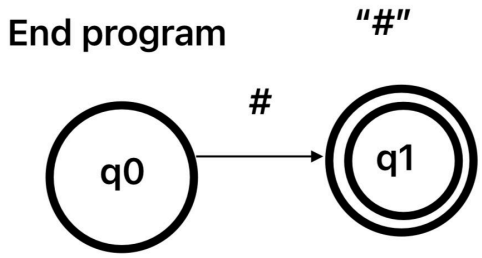
Σ= {UTF-8}

F={q3}

Q0={q0}

	e	n	d
q0	q	qt	qt
q1	qt	q2	qt
q2	qt	qt	q3
q3	qt	qt	qt
qt	qt	qt	qt

End program:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1}

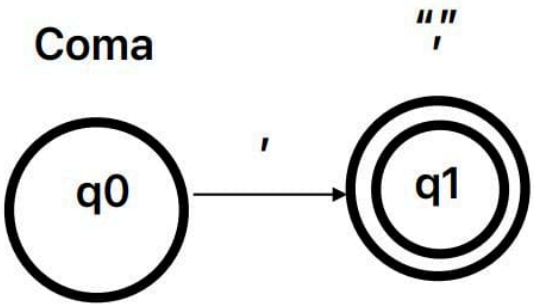
Σ= {UTF-8}

F={q1}

Q0={q0}

	#
q0	q1
q1	qt
qt	qt

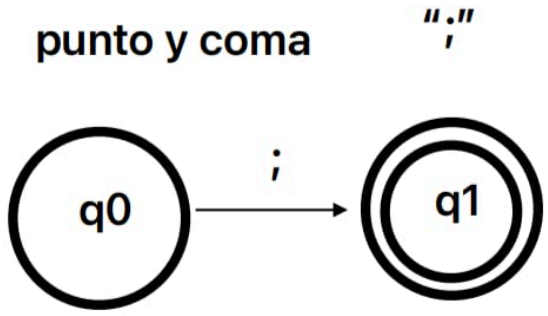
Coma:



A=(VT,VN,Σ,F,Q0) VN= {q0,q1} Σ= {UTF-8} F={q1} Q0={q0}	

	,
q0	q1
q1	qt
qt	qt

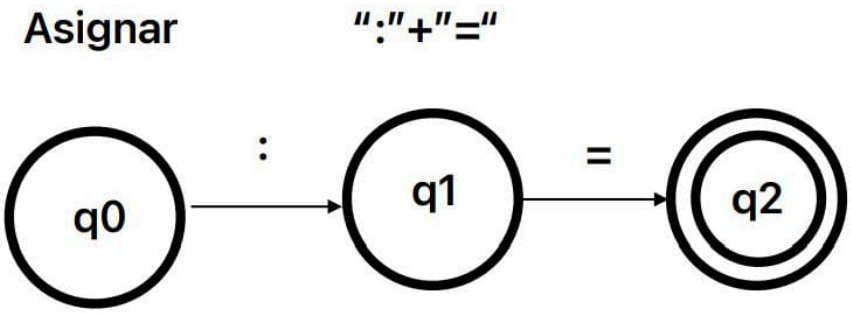
Punto y coma:



A=(VT,VN,Σ,F,Q0) VN= {q0,q1} Σ= {UTF-8} F={q1} Q0={q0}	

	,
q0	q1
q1	qt
qt	qt

Asignacion:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1,q2}

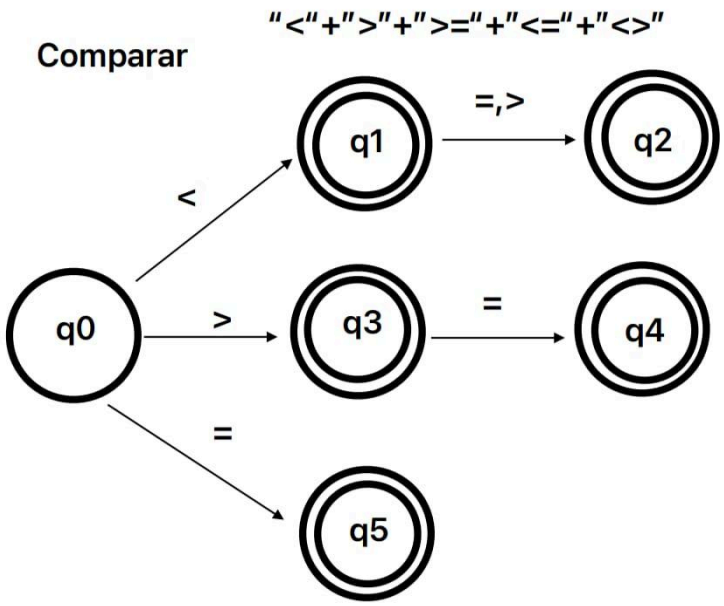
Σ= {UTF-8}

F={q2}

Q0={q0}

	:	=
q0	q1	qt
q1	qt	q2
q2	qt	qt
qt	qt	qt

Comparison:



A=(VT,VN,Σ,F,Q0)

VT= {q0,q1,q2,q3,q4,q5}

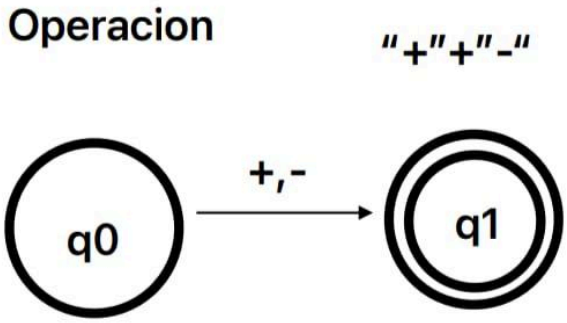
Σ= {<,>,=} U {UTF-8}

F={q1,q2,q3,q4,q5}

Q0={q0}

	<	>	=
q0	q1	q3	q5
q1	qt	q2	q2
q2	qt	qt	qt
q3	qt	qt	q4
q4	qt	qt	qt
q5	qt	qt	qt
qt	qt	qt	qt

Operation:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1}

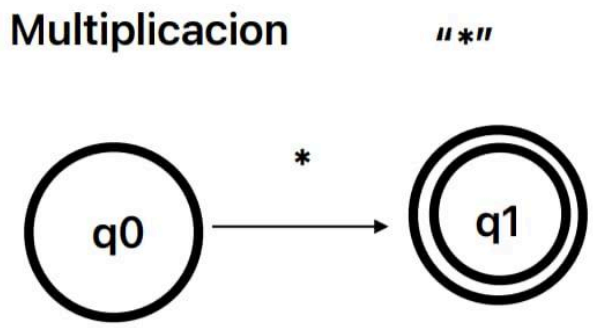
Σ= {UTF-8}

F={q1}

Q0={q0}

	+	-
q0	q1	q1
q1	qt	qt
qt	qt	qt

Multiplicacion:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1}

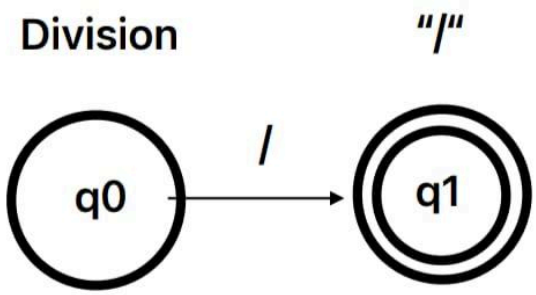
Σ= {UTF-8}

F={q1}

Q0={q0}

	*
q0	q1
q1	qt
qt	qt

Division:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1}

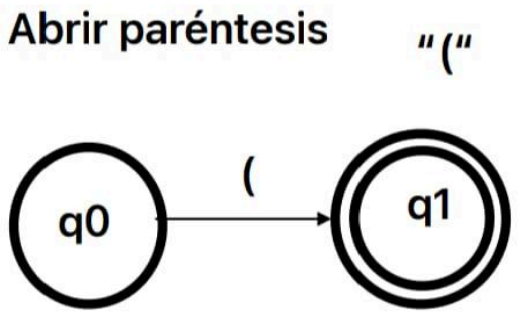
Σ= {UTF-8}

F={q1}

Q0={q0}

	/
q0	q1
q1	qt
qt	qt

Parentesis inicial:



A=(VT,VN,Σ,F,Q0)

VN= {q0,q1}

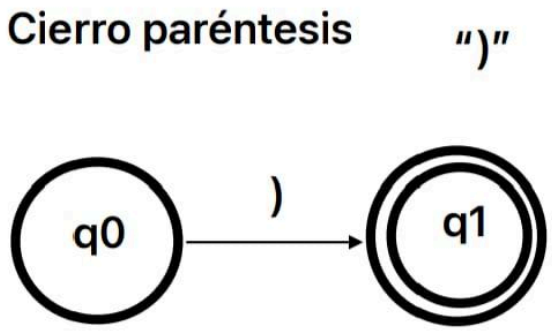
Σ= {UTF-8}

F={q1}

Q0={q0}

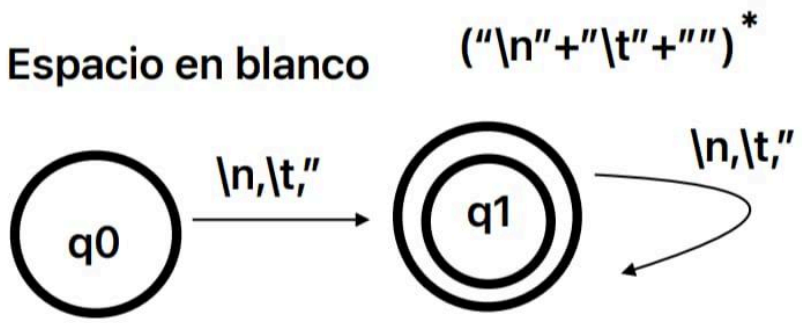
	(
q0	q1
q1	qt
qt	qt

Cierro parentesis:



A=(VT,VN,Σ,F,Q0) VN= {q0,q1} Σ= {UTF-8} F={q1} Q0={q0}		)
	q0	q1
	q1	qt
	qt	qt

Espacio en blanco:



A=(VT,VN,Σ,F,Q0) VN= {q0,q1} Σ= {UTF-8} F={q1} Q0={q0}		\n	\t	´´
	q0	q1	q1	q1
	q1	q1	q1	q1
	qt	qt	qt	qt